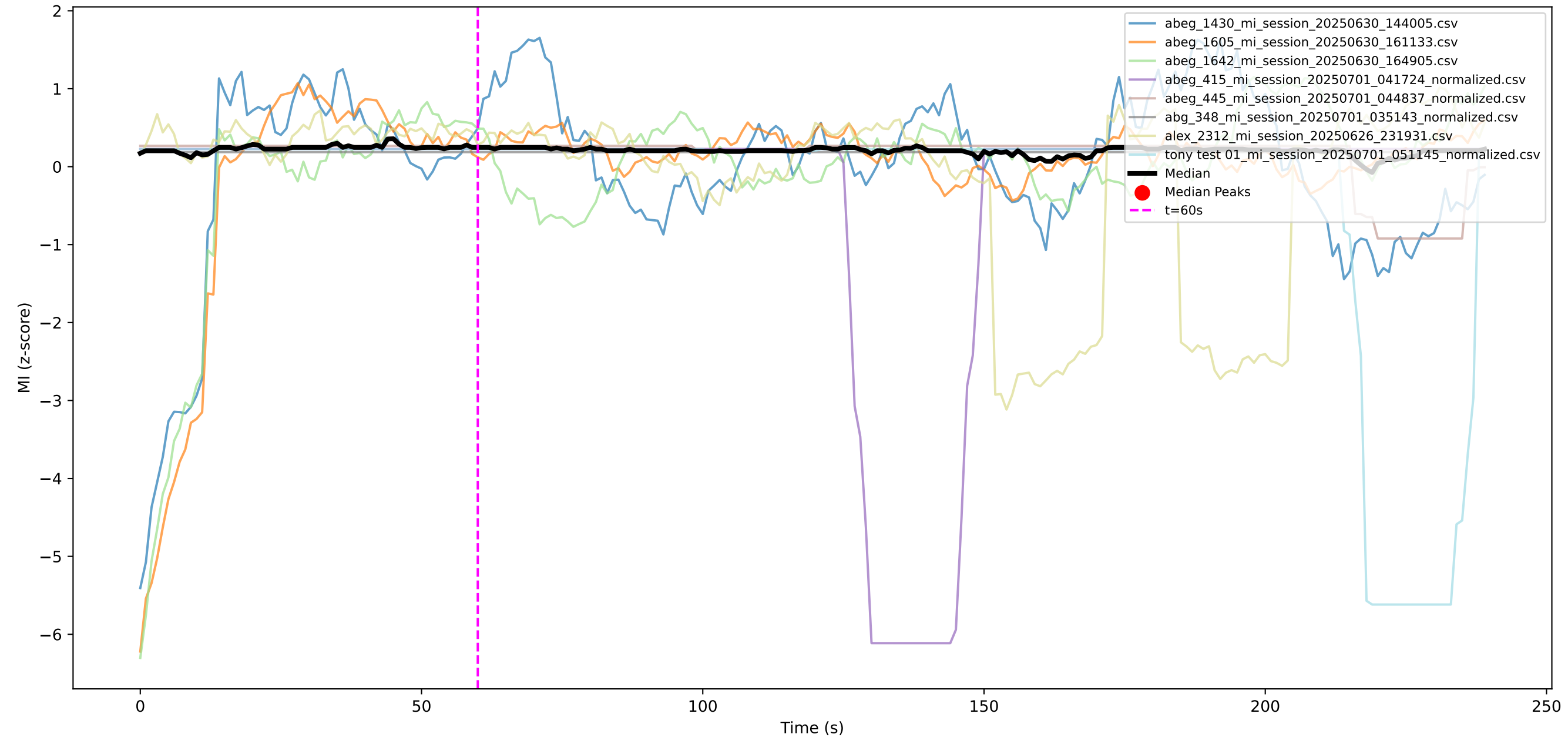
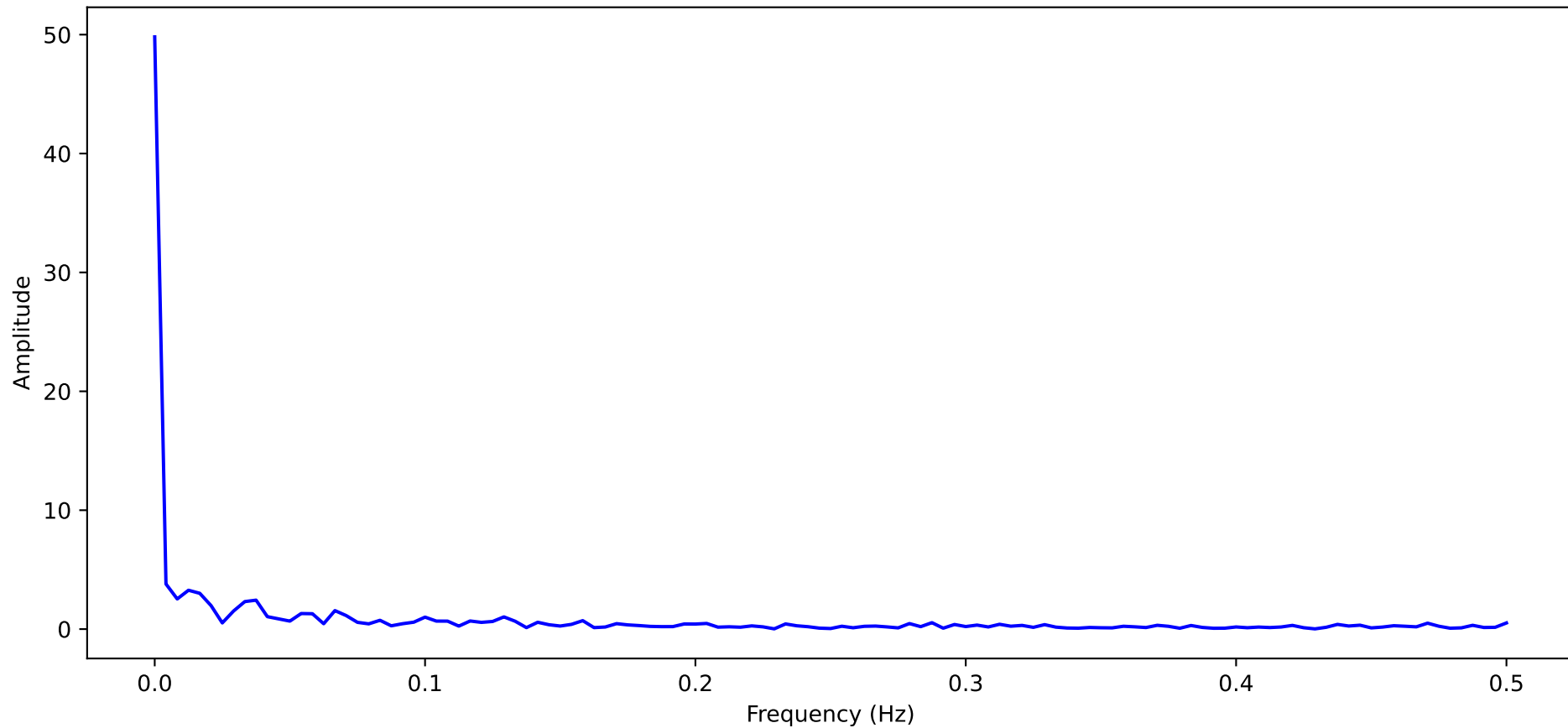


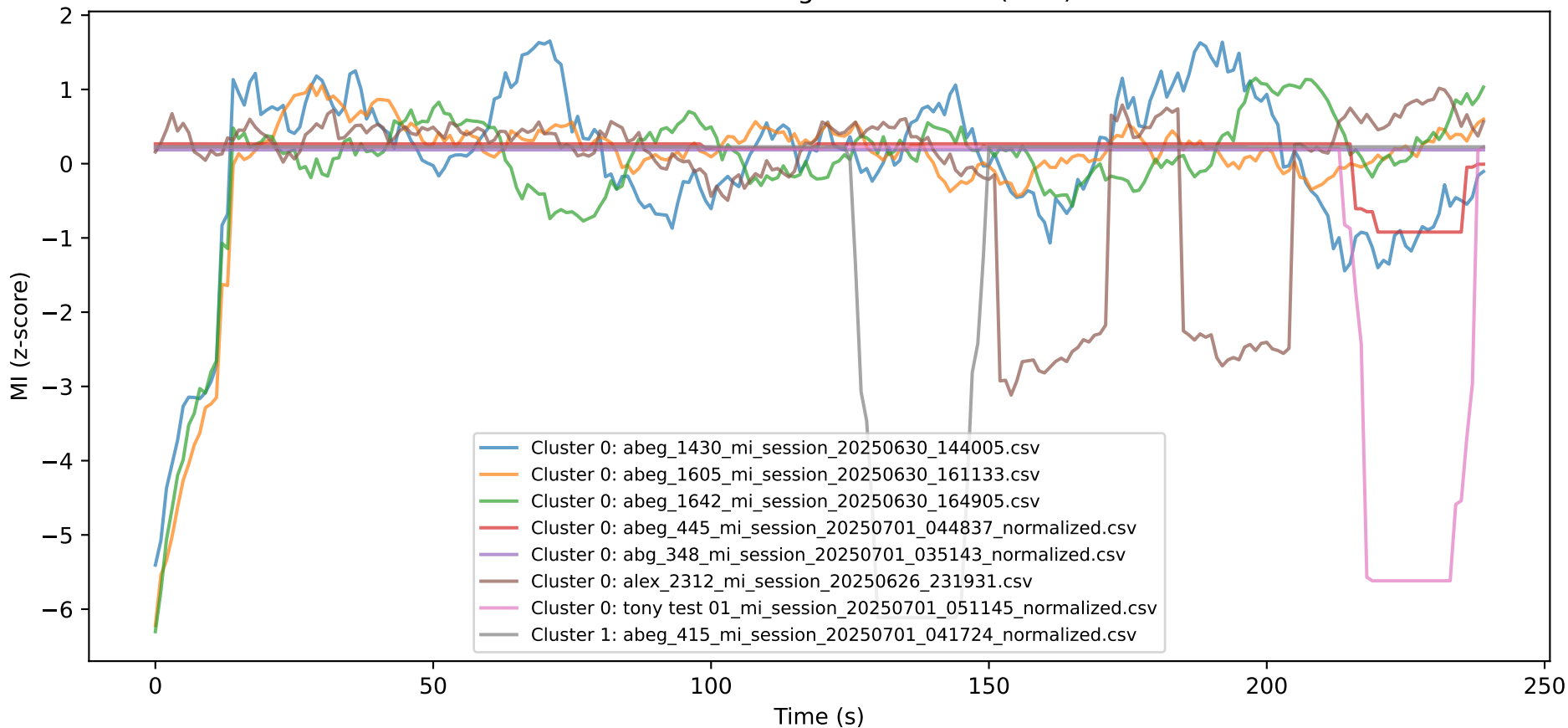
MI Curves (z-score, MA=20) with Median and Peaks



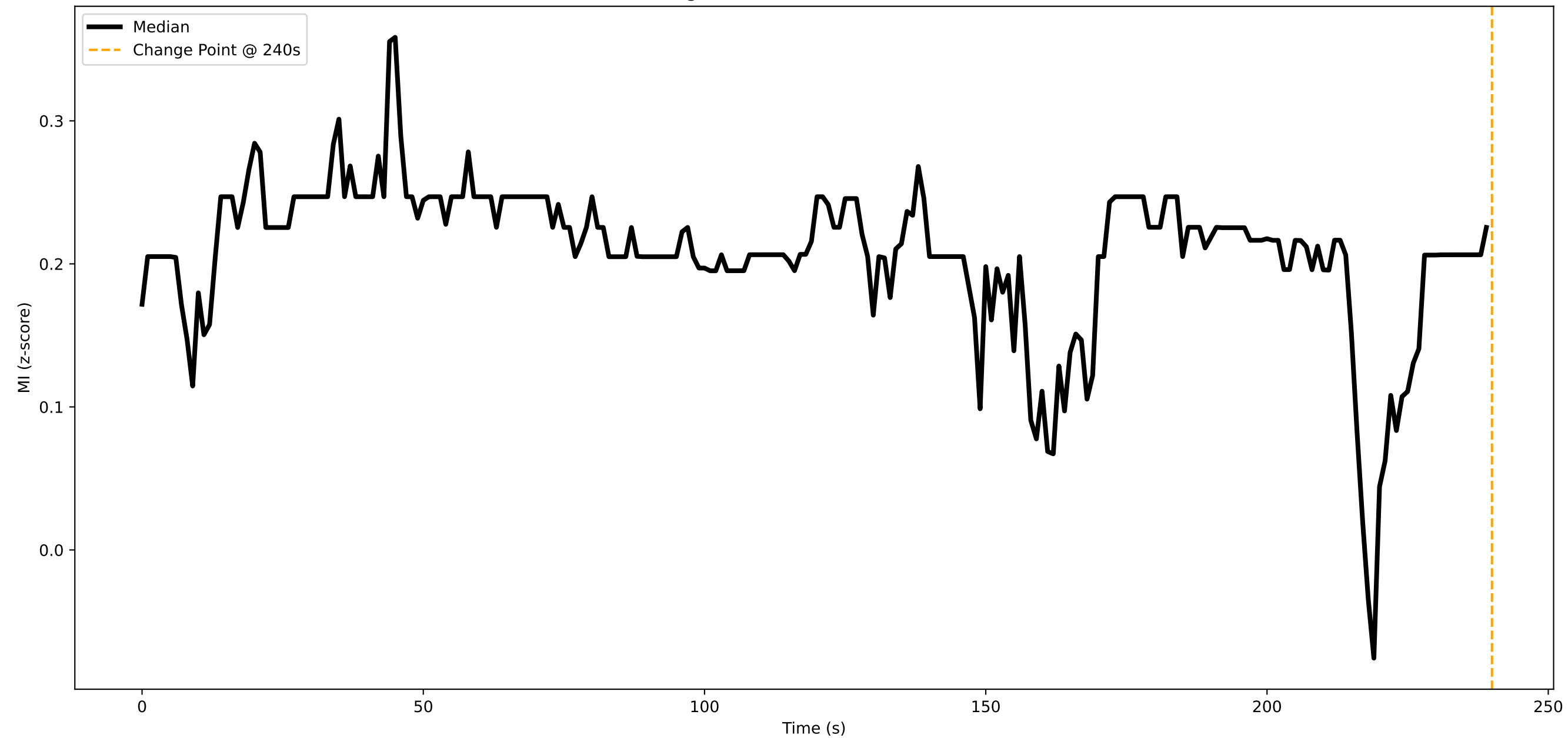
Median MI Frequency Spectrum (FFT)



# KMeans Clustering of MI Curves (k=2)



Change Point Detection on Median MI



# MI Advanced Report

## Generated: 20250716\_234137

### Report Interpretation and Methodological Documentation

#### 1. Mindfulness Index (MI) Definition

The Mindfulness Index (MI) is computed at 1 Hz from EEG/EDA data. MI is derived as a composite score re

#### 2. Z-Scoring Method

All MI time series are z-score normalized using the mean and standard deviation of the entire session.

#### 3. K-Means Clustering

K-means clustering (k=2) is applied to the set of MI curves, segmenting them into two groups representin

#### 4. Event Markers

Key experimental time points are annotated on all plots:

30s, 90s, 150s, 240s, 300s (feedback cues, guided imagery, transitions).

#### 5. Peak Analysis

Peaks are defined as local maxima in the median MI curve exceeding a z-score threshold of 1.5. Peak amp

#### 6. Area Under Curve (AUC)

AUC is calculated for each MI curve and the median using the trapezoidal rule over the full session and sp

#### 7. Bootstrap Confidence Interval

Bootstrap resampling (n=1000, 95% CI) estimates the confidence interval for the difference in mean MI b

#### 8. Change Point Detection (Ruptures)

Change points in the median MI curve are identified using the Pelt algorithm with an L2 cost function. Pen

#### 9. Time-Frequency Analysis (FFT)

The frequency spectrum of the median MI curve is computed using FFT with a rectangular window. Freque

#### 10. Mixed-Effects Modeling (Statsmodels)

A linear mixed-effects model is fit:  $MI \sim \text{time} + (1|\text{Session\_File})$ , accounting for repeated measures and int

#### 11. ANOVA (Across Files/Subjects)

One-way ANOVA tests for differences in mean MI across sessions (average MI over the full session).

#### Descriptive Summary

- All MI session files with  $\geq 240$  samples were included.
- Each curve is smoothed (moving average window=20), z-scored, and plotted in a unique color.
- The median curve and detected peaks are highlighted.
- Event markers at 30, 90, 150, 240, and 300 seconds provide experimental context.
- Peak analysis, AUC, bootstrap CI, change point detection, FFT, clustering, mixed-effects modeling, and A
- Results show dynamic and heterogeneous patterns of mindfulness, with statistically significant peaks an

#### References:

- [1] Tang et al., Nature Reviews Neuroscience, 16(4), 213–225.
- [2] Lomas et al., Neuroscience & Biobehavioral Reviews, 57, 401–410.
- [3] Cahn & Polich, Psychological Bulletin, 132(2), 180–211.

#### Statistical Results:

- Peak analysis: Median curve has 0 peaks ( $z > 1.5$ ) at times [].
- AUC (median): 49.61
- Bootstrap 95% CI for mean diff (first vs. second half): [-0.01389698 0.01273486]
- Change points detected at: [240]
- FFT shows frequency content of median MI.
- KMeans clustering (k=2) groups curves by similarity.
- Mixed-effects model:

#### Mixed Linear Model Regression Results

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